James Akl

Robotics Researcher

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EDUCATION Coursework

PhD	Robotics Engineering	Worcester Polytechnic Institute	GPA 4.00/4.00	**Aug 2019**	**May 2024** (exp.)
BEng	Mechanical Engineering	Lebanese American University	GPA 3.72/4.00	**Sep 2015**	**May 2019**
Minor	Mathematics	Lebanese American University	GPA 4.00/4.00	**Sep 2015**	**May 2018**

EXPERIENCE Details

Doctoral Researcher — Worcester Polytechnic Institute
Manipulation & Environmental Robotics Laboratory — **Aug 2019 – PRESENT**

Develop strategies for metal scrap cutting, waste sorting, and soft robots using robot **vision**, **control**, and **machine learning**.

Teaching Assistant — Worcester Polytechnic Institute
Robotics Engineering Department — **Aug 2019 – May 2020**

Assisted in the **delivery**, **labs**, and **grading** of these courses: *Unified Robotics I, Unified Robotics II, Introduction to Robotics*.

Visiting Researcher — University of Ottawa
Mathematical Modeling & Physical Experiments — **Jun 2018 – Aug 2018**

Modeled the vibrations of ribbed plates, and **experimentally verified** the performance of **analytical** versus **numerical models**.

Research Assistant — Lebanese American University
Multiphysics Modeling & Numerical Simulation — **Nov 2017 – Mar 2019**

Demonstrated the viability of the *ionic buoyancy engine* for underwater propulsion using a coupled multiphysics **numerical model**.

SKILLS Projects

Robotics	Kinematics, Dynamics, Control, Planning, Vision, Learning, Perception, Manipulation, Grasping
Software	C++, Python, Bash, Git, Linux, ROS/Gazebo, MATLAB/Simulink, Mathematica, CMake
Mathematics	Linear algebra, Probability, Statistics, Optimization, Geometric algorithms, Numerical algorithms
Research	Surveying/reviewing, Mathematical modeling, Simulations, Physical experiments, Field work
Publishing	MS Office, LaTeX, Technical writing, Diagramming, Scientific illustration, Video-editing
Social	Presentations, Communication with technical/non-technical audiences, Cross-disciplinary collaboration
Language	English (native), French (professional fluency), Arabic (professional fluency)

RESEARCH Preprints

[2022][1] **Feature-Driven Next View Planning for Robotic Cutting in Metal Scrap Recycling** [Under Review]
[2021][4] **ZeroWaste Dataset: Towards Deformable Object Segmentation in Extreme Clutter** [Under Review]
[2021][1] **Towards Robotic Metal Scrap Cutting: A Novel Workflow and Pipeline for Cutting Path Generation**
[2021][2] **ECNNs: Ensemble Learning Methods for Improving Planar Grasp Quality Estimation**
[2020][1] **Comparing & Optimizing Analytical, Numerical & Experimental Vibration Models for a Simply-Supported Ribbed Plate**
[2019][4] **Simplified Setup for the Vibration Study of Plates with Simply-supported Boundary Conditions**
[2019][1] **Ionic Buoyancy Engines: Finite Element Modeling & Experimental Validation**

[1–4] Superscripts indicate author position in co-authored papers.

AWARDS Proof

Graduated with Distinction, 2019
LAU School of Engineering

LAU Most Novel Club Award, 2018
Founded the Emerging Technologies Club

LAU Engineering Scholarship, 2015
Won a design and programming competition